

Per-Field Categorical G-Code Recoverability from Multi-Modal Sensor Embeddings on a Desktop CNC Mill: A Single-Platform Characterization Study

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Abstract—Controller-resident tampering can alter executed G-code while the loaded file’s hash stays valid. We characterize per-field categorical recoverability of G-code from a desktop CNC mill’s 4 Hz seven-modality sensor stream, using a grammar-constrained six-head Transformer decoder over a frozen denoising encoder under five-fold file-level cross-validation. Categorical fields are recoverable under reference-conditioned (teacher-forced) evaluation, the command head at ~ 0.89 accuracy; measured free-running categorical accuracy collapses to 0.48. The decoder adds +22 pp over the history-free operation-class modal (0.795 vs. 0.572, command-token positions) and $\approx +13$ pp over a history-matched Markov null (0.662); the increment is a 4-of-5-fold effect (at or below the class prior on one fold) and an *upper* bound on sensor-driven recovery. By contrast, generative reconstruction collapses: token accuracy falls $0.78 \rightarrow 0.21$ and digit $0.59 \rightarrow 0.10$ under autoregression, with whole-sequence exact-match at 0.026 under exposure-bias mode collapse, on a corpus in which 77.7% of blocks occupy ≤ 1 sample period. Under open-vocabulary leave-one-class-out evaluation, command recovery collapses to 0.213. Generative coordinate reconstruction is unsupported (source-entropy and sub-Nyquist bounds; companion paper). Tamper detection is deferred to a companion security study; the posture is forensic-support post-hoc audit in a single-platform, closed-world scope, not real-time intrusion detection.

Index Terms—CNC machining, G-code reconstruction, multi-modal sensor fusion, manufacturing cybersecurity, tamper detection, grammar-constrained inference, forensic audit

I. INTRODUCTION

A Computer Numerical Control (CNC) milling machine drives a cutting tool along G-code toolpaths to micrometer tolerances, and every cut leaves a multi-modal signature in vibration, acoustic emission, motor current, and environmental channels. Whether that signature is informative enough to *recover* the executed instruction stream line by line, without controller access, is the question of this paper. The application is forensic-support post-hoc audit, a controller-independent record of executed instructions, not real-time intrusion detection [1]–[4]. All results are conditional on a single desktop mill (air-cut and UHMW-PE active-cut runs, 4 Hz envelope-extracted stream); cross-platform replication is future work (Section VIII).

CNC milling is a harder signal environment than the additive-manufacturing side-channel analog [5]–[7]: simultaneous multi-axis interpolation, nonlinear force engagement, and frame-borne vibration corrupt the signal, and the sensor-to-instruction mapping is many-to-one [8], [9]. We take a prior multi-modal denoising encoder (96.3% nine-class operation accuracy [10]) as a frozen feature extractor and a companion grammar tokenization [11] as vocabulary, and ask what its per-window latent carries about the row-level instructions. Three task-specific obstacles shape the design: a 64-second window straddles many G-code rows, so window-level targets destroy row information; the numeric vocabulary holds thousands of sparse bucket tokens; and positional metadata is a shortcut that inflates accuracy without proving sensor-driven recovery [12].

a) Headline findings (stated as bounds).: The central quantities are not the closed-vocabulary ~ 0.89 command accuracy but two bounds. Above the operation-class modal predictor, the decoder adds a +22 pp class-prior-controlled within-class command increment, an *upper* bound on sensor-driven recovery that is a 4-of-5-fold effect, at or below its class prior on one fold, and $\approx +13$ pp against a history-matched sensor-free Markov null (Section VII-A0a). On an open-vocabulary leave-one-class-out holdout, command recovery collapses to a floor of 0.213. End-to-end generative reconstruction is at floor (autoregressive whole-sequence exact-match 0.026). We reserve “recovery” for the per-field categorical heads and treat generative coordinate reconstruction as unsupported on this stack. Because 77.7% of rows occupy ≤ 1 sample period at 4 Hz, all recovery claims are window-aggregated row-modal, not sub-sample per-row.

b) Contributions.:

- (1) **Per-row and full-window supervised targets.** A per-row formulation maps each sensor window to the single G-code line firing within it ($\sim 19,500$ samples per fold versus ~ 300 at window level), removing within-window label ambiguity; the headline configuration trains on the full-window variant, which restores within-window row identifiability (Section VI-F).
- (2) **G-Code Decoder architecture.** A Transformer decoder

with six structured heads (token, type, motion-command, parameter-type, sign, digit) over the frozen encoder latent [10], grammar-constrained so structurally invalid transitions are masked at runtime [13], [14].

- (3) **Per-axis recoverability characterization** reporting, per field, axis detection, motion-direction correctness, and numeric recovery over X, Y, Z, R.
- (4) **Ablation matrix** over metadata access, per-row vs. full-window targets, seven sensor modalities, and a no-numeric retraining.
- (5) **Cross-class generalization study**: leave-one-class-out across nine operation classes, testing memorized per-class shortcuts against within-envelope generalization.
- (6) **From-scratch Transformer seq2seq baseline** scoping the architectural lift; a companion limit paper [15] accounts for why the recoverability pattern takes this shape.

II. RELATED WORK

Sensor-based machining monitoring is mature but coarse-grained: force, vibration, acoustic emission, temperature, and motor current support tool-condition and process assessment [8], [16], [17], via deep feature learning [9], [18] and multi-modal fusion [19]–[21], but at session- or class-level targets [22]–[24]. Grammar-constrained decoding prunes structurally invalid tokens at inference [13], [14], [25], [26], and G-code fits this treatment [11], [27]. The closest analog is additive-manufacturing side-channel toolpath reconstruction from acoustic [5], [6], optical [28], and acoustic-magnetic [29] channels, but at kHz–MHz raw rates rather than our 4 Hz audio-RMS envelope; Kimmell *et al.* [4] perform binary CNC attack detection, not reconstruction. File-hash signing [30] and controller attestation [31] are orthogonal non-sensor defenses. To our knowledge this is the first per-instruction, per-field categorical recovery of CNC *milling* G-code from a multi-modal sensor stream; the 5-fold file-level cross-validation and nine-class LOCO protocol support that claim rather than standing as a co-equal novelty leg.

III. PROBLEM FORMULATION

A. Task Definition

Let $W \in \mathbb{R}^{T \times C}$ be a sensor window of T samples over C channels aligned to a CNC milling operation, and g the G-code line executing within it. We learn a mapping

$$f_\theta : W \mapsto \hat{g}, \quad \hat{g} = (\hat{t}_1, \hat{t}_2, \dots, \hat{t}_L),$$

producing a length- L token sequence from a vocabulary $\mathcal{V}$. Training minimizes a multi-head cross-entropy loss with the structured heads of Section IV-C. At inference the decoder generates tokens autoregressively under a grammar mask that zeros structurally invalid transitions in the legacy-token head.

B. Vocabulary and Token Types

Each motion block is a *rapid move* (G0, traverse) or a *cutting move* (G1/G2/G3, linear and arc feeds); all blocks

are tokenized by a hybrid scheme that separates structural primitives from numeric values [11], [27]. The vocabulary has four classes: SPECIAL (PAD, BOS, EOS, UNK, MASK); COMMAND (literal motion or machine commands G0, M3, ...); PARAM (axis or rate address letters X, Y, Z, F, S, ...); and NUMERIC (bucketed values, one bucket per (axis, magnitude) pair). NUMERIC dominates, with 1,048 NUM_X, 1,086 NUM_Y, and 231 NUM_Z buckets; total vocabulary size 2,418. For example, G1 X1.4919 Y1.4848 F22. becomes BOS G1 X NUM_X_1492 Y NUM_Y_1485 F NUM_F_22 EOS.

C. Per-Row Target Formulation

A 256-sample window at 4 Hz spans 64 seconds and averages 60 G-code rows, so a window-level target aggregates across rows and discards row-level distinctions such as an X positive versus negative move. We instead emit one (W, g_r) pair per row r firing within W , with g_r that row’s token sequence. This removes within-window label ambiguity, raises the training-sample count by an order of magnitude ($\sim 19,500$ versus ~ 300 per fold), and blocks positional-shortcut inference on `(window_index, source_file)`. A complementary *full-window* formulation keeps one sample per window with a multi-line target of all fired rows in temporal order. The headline configuration (Table I) trains on full-window targets, which restore within-window row identifiability (Section VI-F); per-row targets remain the preferred autoregressive deployment mode, since per-row predictions are short and independent across rows, with the command-head cost of the per-row mode given by the 0.499-versus-0.888 target-mode contrast (Section VI-F).

D. Evaluation Metrics

We report seven metrics: token accuracy (the headline; fraction of non-padding positions whose argmax matches the target), sequence accuracy (strict exact-match over all non-padding tokens in a sample), and per-token accuracy on the type (4-way), motion-command (6-way), parameter-type (10-way), sign (3-way), and per-digit NUMERIC heads. Macro precision, recall, and F1 follow per head [32].

a) Teacher-forced vs. autoregressive evaluation.:

Teacher-forced (TF) evaluation supplies the ground-truth previous token at every position (the training objective); *autoregressive* (AR) evaluation consumes the model’s own predictions, measuring end-to-end generation when no controller log anchors the sequence. The categorical heads (type, command, parameter-type, sign) are evaluated from the ground-truth-conditioned forward pass in both regimes and therefore carry a single number each; because they read the decoder hidden state, which is computed from the conditioning token stream (Section IV-C1), their free-running accuracy is distinct and collapses when measured (command 0.483 ± 0.049). The autoregressive-token heads score lower under AR because exposure-bias errors compound. The headline table reports both regimes for the token heads.

E. Threat Model (Companion Study)

The application motivating per-field recoverability is out-of-band verification of the executed instruction stream against controller-resident tampering. The full threat model — trusted and untrusted elements, the attack-class taxonomy, and per-class detectability — is developed in a companion security study (in preparation); the present paper characterizes only which G-code fields are recoverable, the precondition for any such monitor.

IV. METHOD

A. System Overview

The system has three stages (Figure 1): a frozen multi-modal encoder produces a memory sequence from the sensor window; a Transformer decoder generates the executing G-code line via cross-attention; and a grammar mask zeros structurally invalid softmax transitions at inference. Encoder training is described in [10].

B. Sensor Encoder (Frozen)

The frozen encoder (MM-DTAE-LSTM) is a multi-modal denoising autoencoder over seven modality groups totaling $C = 98$ channels, fused by learned modality-weighting gates [19], [20] and a multi-head self-attention stack [33], with a two-layer LSTM [34], [35] producing the temporal memory. Its ≈ 5.1 M active parameters reach 96.3% test-set accuracy on the nine-class operation-type task [10], trained on the same file-level 5-fold splits.

C. Decoder Architecture

G-Code Decoder is an 8-layer Transformer decoder with $d_{\text{model}} = 384$ and $h = 12$ attention heads per layer, each applying masked self-attention, cross-attention over the encoder memory, a GELU feed-forward network, and pre-layer-norm residuals [36] with dropout [37] 0.1. It has 31.28 M trainable parameters; with the frozen encoder (≈ 5.1 M active) the total system is ≈ 36.4 M.

1) *Multi-Head Structured Output*: Six prediction heads share the decoder’s final hidden state at each position, each a linear projection then softmax, trained jointly with a weighted cross-entropy loss:

$$\mathcal{L}_{\text{decoder}} = \lambda_{\text{tok}}\mathcal{L}_{\text{tok}} + \lambda_{\text{type}}\mathcal{L}_{\text{type}} + \lambda_{\text{cmd}}\mathcal{L}_{\text{cmd}} \\ + \lambda_{\text{param}}\mathcal{L}_{\text{param}} + \lambda_{\text{digit}}\mathcal{L}_{\text{digit}} + \lambda_{\text{sign}}\mathcal{L}_{\text{sign}}.$$

Default weights are $\lambda_{\text{tok}} = 3.0$ with all auxiliary $\lambda = 1.0$; the auxiliary heads regularize the shared representation toward token-type-aware features. The heads are: the legacy token head, a vocabulary-sized softmax (2,418 classes) that is the primary inference output, grammar-masked (Section IV-C2) before the softmax; a 4-way type head over {SPECIAL, COMMAND, PARAM, NUMERIC}; a 6-way motion-command head over {G0, G1, G2, G3, G53, M30}; a 10-way parameter-type head over {X, Y, Z, I, J, K, R, F, S, P}; a 3-way sign head over {POSITIVE, NEGATIVE, ABSENT}; and a six-position 11-way digit head reconstructing each NUM token digit-by-digit. The categorical heads read this shared hidden

state, which is computed from the conditioning token stream; the headline categorical values condition on the ground-truth stream, and under free-running generated-stream conditioning the heads collapse (command 0.483 ± 0.049 ; Section III-D).

2) *Grammar-Constrained Decoding*: G-code follows a regular grammar. We encode the local rules as a static $|V| \times |V|$ binary transition mask on the legacy-token logits at each inference step, cutting the valid next-token set from $|V| = 2,418$ to 31.8 tokens on average, following established runtime grammar-mask patterns [13], [14], [25], [26]. A separate inference-time FSM layer tracks the most recent move-command and forbids command-state violations the bigram mask cannot express (e.g. R/I/J after G0/G1). With the FSM active, command-state (V1) violations drop to 0 (100% reduction) at a 5-fold accuracy cost within fold-to-fold noise (paired- t and Wilcoxon $p > 0.4$); we adopt it as the default deployment-time configuration. Full counts are in `audit/fsm_vs_bigram_paired_stats.json`.

3) *Scheduled Sampling*: Teacher forcing is the training default; the headline configuration uses scheduled sampling [38] with $p_{\text{max}} = 0.5$, which lifts command accuracy +17 pp over the $p_{\text{max}} = 0$ teacher-forced variant. Because the sweep that selected it touches held-out test metrics, this is a configuration choice, not an independent generalization estimate.

D. Training Configuration

The decoder is optimized with AdamW [39] ($\eta = 5 \times 10^{-5}$, weight decay 0.05) for up to 300 epochs with early stopping on validation token accuracy (patience 75), a 10-epoch linear warm-up, batch size 64, and gradient clipping at norm 1.0. The baseline trains *without* positional metadata (Section VI-F ablates this).

V. EXPERIMENTAL SETUP

A. Dataset

The corpus is 120 aligned CSV files from a Bantam Tools desktop CNC mill [10]. Six multi-sensor boards are aligned at an effective 4 Hz rate across seven modalities (accelerometer, gyroscope, magnetometer, RGBA color, temperature [the ablations’ “environmental”], audio RMS, motor current [“electrical”]); after the encoder paper’s consistency audit we retain $C = 98$ channels, each envelope-extracted to 4 Hz by a single-pole IIR filter that trades kHz-band acoustic-emission content [8] for a ~ 1.5 kB/s stream. The nine operation classes cross three motion strategies (face, pocket, adaptive) with three run conditions [10]: standard air-cuts, 150025 active cuts engaging UHMW-PE at 0.635 mm depth, and damaged-spindle air-cuts (one drive band removed; `audit/operation_class_inventory.json`). All runs execute dry on one low-amplitude, stepper-driven platform; whether the factorization transfers to a production VMC is open.

B. Preprocessing and Windowing

Sliding windows of $T = 256$ samples (64 s at 4 Hz) at stride 64 are aligned to their overlapping G-code rows. The per-row target formulation emits one (W, g_r) pair per row firing

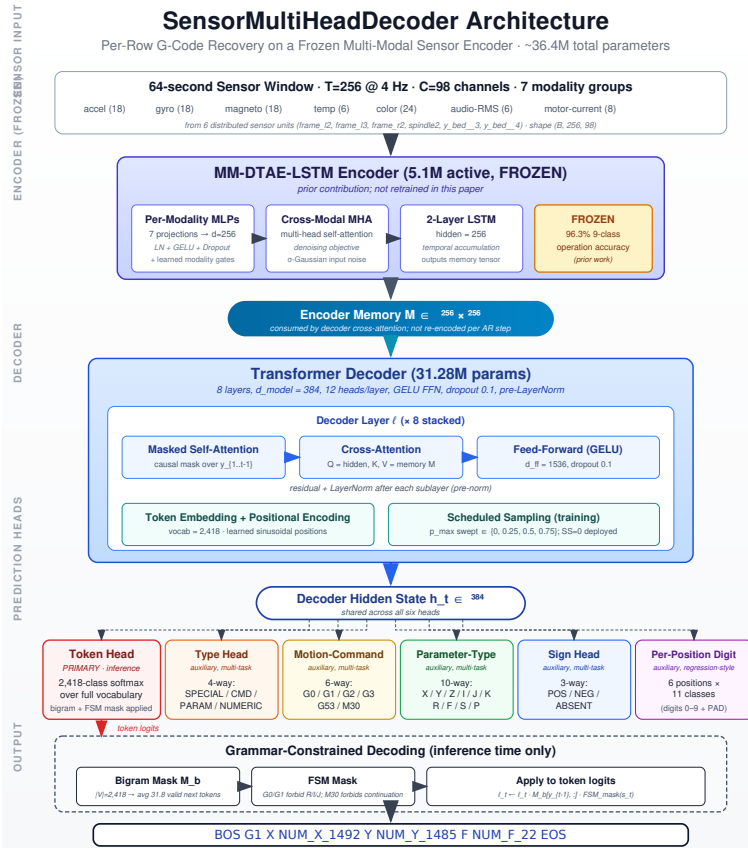


Fig. 1: System overview. A 64-second sensor window (256 samples $\times$ 98 channels across seven modality groups) is mapped by the frozen MM-DTAE-LSTM encoder (≈ 5.1 M active parameters) to a 256×256 memory tensor. An 8-layer Transformer decoder ($d_{\text{model}}=384$, 12 heads) consumes this memory via cross-attention and generates tokens autoregressively. Six prediction heads share the decoder’s hidden state and provide structured output (token, type, motion-command, parameter-type, sign, per-position digit). A static grammar mask zeros structurally invalid token transitions in the legacy-token softmax at inference time.

within window W , yielding $\sim 19,500$ training pairs per fold; the headline configuration trains on the full-window variant (Section III-C). The 4 Hz rate constrains recoverability. A row-execution audit (audit/gcode_row_durations.json) shows 77.7% of row instances occupy ≤ 1 sample period, against the ≥ 10 kHz of conventional tool-condition monitoring [8].

C. Five-Fold File-Level Stratified Cross-Validation

File-level stratified five-fold cross-validation prevents within-file leakage, and each fold covers all nine classes. A G-code-line coverage-repair step swaps files between train and test until every test line also appears in training, constructing a closed-vocabulary regime (97.6% of distinct test lines, 99.0% of instances seen verbatim). The headline numbers are therefore within-vocabulary classification, not open-vocabulary recovery; the matched open-vocabulary number is the leave-one-class-out (LOCO) command accuracy of 0.213 ± 0.191 (Section VI-H). Overlapping windows (75% overlap) make adjacent within-file windows non-independent, so we take the per-fold mean as the unit of analysis.

D. Statistical Analysis

For each five-fold ablation we report the cross-fold mean and standard deviation, with $n = 5$ per group. All p -values are *descriptive* and the bootstrap 95% intervals a smoothed, not coverage-calibrated, indication of cross-fold spread; conclusions rest on effect sizes large relative to that spread rather than on significance thresholds, with nonparametric (Wilcoxon) checks accompanying the paired tests where reported. Single-fold pilots are excluded from quantitative claims; the leave-one-class-out protocol (Section VI-H), though single-fold per holdout, is reported as a finite-population descriptive result over the nine-class corpus.

E. Software and Compute

The decoder is implemented in PyTorch and trained on NVIDIA RTX A6000 GPUs; preprocessing uses scikit-learn [32] and SciPy [40], with full code and configuration released with the manuscript.

VI. RESULTS

A. Non-Neural Baseline Comparison

Two lightweight classifiers on the mean-pooled encoder embedding the decoder consumes, a scikit-learn HistGradientBoosting (HGB) tree-boosters and a two-layer MLP probe, hit the class-prior ceiling on the binary fields (≈ 0.88 *has-X*, 0.99 *has-F*) but fall to macro-F1 0.23–0.25 on multi-class command identity; HGB reaches macro-F1 0.86 on the one within-window-diverse field *has-Z*, so the pooled embedding carries some per-row Z signal (fold-1 reference on per-row targets; `audit/nonneural_baselines.json`).

B. Headline Accuracy and Per-Field Recoverability

Table I reports headline accuracies per head across five folds under teacher-forced (TF) and autoregressive (AR) evaluation. The categorical heads, each reported from the ground-truth-conditioned pass (free-running they collapse, command 0.483; Section III-D), reach type 0.9838, command 0.8875 ± 0.0558 , parameter-type 0.9931, and pooled motion-sign 0.9888. The autoregressive-token heads diverge sharply: token accuracy 0.7811 (TF) versus 0.2148 (AR), numeric 0.5850 (TF) versus 0.1038 (AR). Sequence exact-match, defined only under AR, is 0.0263 ± 0.0185 , fewer than 3% of windows reproduced verbatim; it is the deployment-relevant generative figure. The TF–AR gap is the exposure-bias signature of autoregressive mode-collapse (analyzed in the extended version of this work), and the headline is seed-robust (four seeds span 0.0085, Appendix C). A float-epsilon artifact at the finest decimal slot was corrected by a round-encoding retrain; headline and categorical metrics drift ≤ 1.5 pp under it, so Table I uses the released checkpoint and the patched retrain serves only the numeric tier (Section VI-D).

We reserve *recovery* for the per-field categorical heads (command, parameter-type, sign) under reference-conditioned evaluation. End-to-end generative reconstruction is at floor, so this paper claims per-field categorical recoverability, not toolpath reconstruction. Command macro-F1 is 0.879, moderately below accuracy on the long-tailed class distribution, with recovery across all six classes and no collapse (lowest G2 at 0.82; per-class F1 in the extended version); parameter-type macro-F1 is 0.986.

Three caveats bound these numbers. They are closed-vocabulary, in-distribution (97.6%/99.0% verbatim test-line overlap), so command 0.8875 is within-vocabulary classification, not an open-vocabulary deployment number. The frozen encoder shared the file-level fold partitions, so the headline carries a representation-reuse component bounded in Appendix A. The LOCO cross-class command accuracy of 0.213 ± 0.191 (Section VI-H) is the open-vocabulary floor.

C. Per-Axis Recoverability

For each axis letter we score *has-axis* (axis present in the predicted block), *sign* (motion sign when present), and *value MAE* (mean absolute error where both blocks carry the axis); near-zero-presence fields cannot be meaningfully evaluated (Table II).

Four patterns emerge. Axis-presence detection is uniformly high: X, Y, Z at 1.000 ± 0.000 , R at 0.994, and F at 0.960 on thin support. Motion-sign recovery is axis-asymmetric but *prior-driven*, not a sensor-separability ranking: Y-sign (0.997) and X-sign (0.942) ride near-deterministic priors, while Z-sign (0.938), the only balanced-distribution sensor-bearing axis, scores lower as a harder sensor-driven binary classification, not weaker coupling. Value MAE is small for Z (0.009 patched) and arc-radius R (0.021) but larger for X (0.248) and Y (0.205), all in inches, localized to middle digits by the per-digit decomposition. Spindle (S) and arc parameters (I, J) have effectively zero positive support and cannot be evaluated (Figure 2).

D. Numeric Digit Head Decomposition

Per-slot TF accuracy on the corrected retrain is bathtub-shaped (Figure 2b): slot 0 (tens) at ~ 1.00 descends to the *true precision floor* at slot 4 (0.42, thousandths, the finest digit the V8 tokenizer retains [15]). Slot 5 returns to 0.99 because the V8 tokenizer (the hybrid vocabulary of Section III-B) quantizes X/Y/Z onto a 0.001 grid, making the tokenized ten-thousandths digit structurally near-zero entropy, deterministically reproduced rather than information-bearing [15]. The decoder therefore behaves as a coarse-resolution coordinate recoverer, tracking the corpus per-(axis, slot) source-entropy ceiling (corrected pooled Pearson -0.89).

E. Per-Operation-Class Recoverability

Autoregressive predictions stratify sharply by operation class from a single mechanistic cause: *face* reaches token 0.56 / numeric 0.40 while every other class sits in a narrow 0.09–0.25 token band, because the corpus-modal AR output the decoder collapses to *is* a facing-pattern toolpath, so *face* scores by coincidental agreement, not per-class capability (per-class figure in the extended version).

F. Metadata, Target-Mode, and Numeric-Collapse Ablations (Extended Version)

Three ablations refine the headline and are reported in full in the extended journal version of this work (in preparation): positional metadata does not improve the categorical heads under any regime but mitigates autoregressive mode-collapse (+7.5–8.0 pp on the AR token/numeric heads); the full-window target formulation lifts command accuracy from per-row 0.499 to 0.888 by restoring within-window row identifiability; and a no-numeric retraining decomposes the AR collapse into a numeric-desynchronization component and a dominant intrinsic structural-stream collapse.

G. Sensor-Modality Ablation

For each of seven sensor modality groups (accelerometer, gyroscope, magnetometer, environmental, color, audio-RMS, electrical) we zero the corresponding channels at the encoder input and evaluate without retraining, a *lower bound* on each modality’s value. Every modality is critical to the command head at a statistically uniform magnitude: zeroing any single

TABLE I: Headline test-set metrics for the G-Code Decoder under 5-fold file-level cross-validation, full-window targets, no positional metadata exposed. Mean $\pm$ std over 5 folds. Two evaluation regimes are reported: teacher-forced (TF), where each step receives the ground-truth previous token, and autoregressive (AR, greedy / beam-width 1), where the decoder consumes its own previous predictions. The categorical heads — type, command, parameter-type, sign — report a single reference-conditioned value (the AR column is marked ‘–’); measured free-running categorical accuracy is reported in the footnote. Only the autoregressive-token heads (token, sequence, numeric) carry distinct TF and AR values.

Head	TF Accuracy	AR Accuracy	Macro Precision	Macro F1
Token	0.7811 $\pm$ 0.0216	0.2148 $\pm$ 0.0685	0.4099 $\pm$ 0.0406	0.3899 $\pm$ 0.0405
Sequence [†]	–	0.0263 $\pm$ 0.0185	–	–
Type	0.9838 $\pm$ 0.0085	–	0.9091 $\pm$ 0.0352	0.8833 $\pm$ 0.0451
Command	0.8875 $\pm$ 0.0558	–	0.9342 $\pm$ 0.0407	0.8790 $\pm$ 0.1084
Param-type	0.9931 $\pm$ 0.0036	–	0.9914 $\pm$ 0.0073	0.9857 $\pm$ 0.0130
Sign	0.9888 $\pm$ 0.0063	–	0.9658 $\pm$ 0.0215	0.9724 $\pm$ 0.0157
Numeric (overall digit)	0.5850 $\pm$ 0.0331	0.1038 $\pm$ 0.0322	–	–

Macro precision and F1 are unweighted means over all observed classes; the categorical heads carry a single (reference-conditioned) value. Numeric is the digit-head accuracy pooled across all six digit positions; the per-position bathtub is characterized in Section VI-D. The TF / AR gap on the token and numeric heads is the exposure-bias signature; the categorical AR column is marked ‘–’ because those values are reference-conditioned; measured free-running categorical accuracy collapses (command 0.483 $\pm$ 0.049, type 0.551 $\pm$ 0.087, parameter-type 0.533 $\pm$ 0.046; audit/ar_categorical_freerunning.json). [†] Sequence exact-match is reported only under AR; a sequence-level TF score is not separately well-defined (see Section VI-B).

TABLE II: Per-axis G-code recoverability across 5 folds (mean $\pm$ std). **Value-MAE column re-derived on the float-epsilon-corrected retrain** (checkpoints/full_window_5fold_patched/; audit/patched_numeric_analysis.json); the categorical (has-axis, sign) columns are checkpoint-invariant. *has-axis Acc / F1* are the classification metrics for whether the axis is present in the decoded text; *sign* is the direction of motion (POSITIVE / NEGATIVE / ABSENT); *value MAE* is the mean absolute error of the numeric value, computed only on samples where both the predicted and true G-code carry the axis. The headline recoverability claim is restricted to the four well-supported axes **X, Y, Z, R**. The F row ([†]) is reported for the *full-window regime only*, where F appears in 22% of windows; in the per-row regime F support is only 0.6% and is not evaluable. I, J, S have effectively zero positive support in both regimes and are not evaluable. The per-axis sign-accuracy asymmetry (Y 0.997 vs. X 0.942 vs. Z 0.938) is *prior-driven*: Z is the only sign-bearing axis whose sign carries non-trivial classification load ($H_Z^{\text{sign}}=1.04$ bits), whereas Y’s sign is prior-deterministic ($H_Y^{\text{sign}}=0.08$ bits) and X’s is near-deterministic ($H_X^{\text{sign}}=0.38$ bits); the per-axis sign source-entropy decomposition is developed in the companion limit paper [15], so the high Y/X sign accuracy reflects the prior, not sensor discrimination.

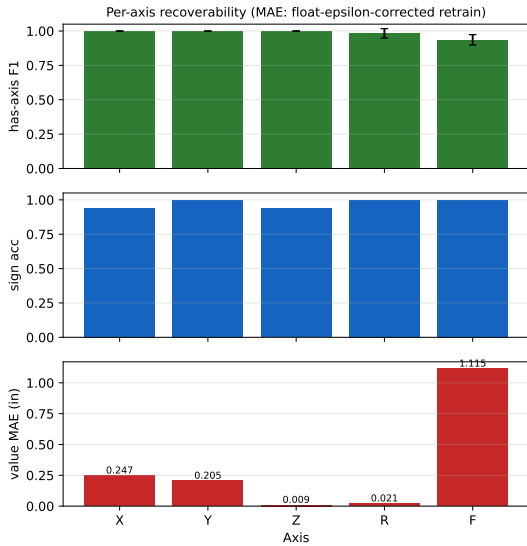
Axis	has-axis Acc	has-axis F1	Sign Acc	Value MAE	Presence Recall
X	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.942 $\pm$ 0.033	0.248 $\pm$ 0.140	1.000 $\pm$ 0.000
Y	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.997 $\pm$ 0.006	0.205 $\pm$ 0.080	1.000 $\pm$ 0.000
Z	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	0.938 $\pm$ 0.023	0.009 $\pm$ 0.006	1.000 $\pm$ 0.000
R	0.994 $\pm$ 0.012	0.983 $\pm$ 0.034	1.000 $\pm$ 0.000	0.021 $\pm$ 0.011	1.000 $\pm$ 0.000
F [†]	0.960 $\pm$ 0.023	0.936 $\pm$ 0.038	1.000 $\pm$ 0.000	1.115 $\pm$ 0.350 (inch/min)	0.851 $\pm$ 0.093
S, I, J	insufficient support	–	–	–	–

X, Y, Z, R presence is fully recovered. The sign asymmetry is prior-driven, not a sensor-separability ranking: Y (0.997) and X (0.942) sign accuracy reflect their near-deterministic sign priors ($H_Y^{\text{sign}}=0.08$, $H_X^{\text{sign}}=0.38$ bits), while Z (0.938) is the only axis whose sign is genuinely balanced ($H_Z^{\text{sign}}=1.04$ bits) and therefore the only one carrying real sensor-driven sign signal. Value MAE is small for Z (0.009) and R (0.021) and larger for X/Y (0.21–0.25), all in inches (the corpus’s native unit; the platform runs in inch mode). Against the per-axis coordinate span over the corpus ($X \approx 3.8$, $Y \approx 3.8$, $Z \approx 0.7$, $R \approx 7.9$ in) these MAEs are $\approx 6.5\%$ and 5.4% of span on the large planar X/Y axes but only $\approx 1.2\%$ and 0.3% on the shallow-depth Z and arc-radius R axes — coordinate value is recovered to coarse resolution on X/Y and near-exactly on Z/R. These are the patched-retrain values (Z is 0.009 patched vs. 0.012 on the released checkpoint recomputed identically; an earlier reported 0.042 was a pre-patch analysis-pipeline artifact, not a checkpoint difference). The F-row MAE is on a different scale because feed-rate values themselves are larger (four distinct posted values, 7.3–40 in/min, in this corpus).

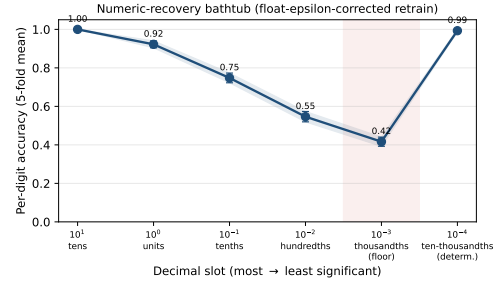
group drops command accuracy from the full-modality baseline 0.888 ± 0.056 to the 0.449–0.484 band, a uniform –40 to –44 pp collapse. The seven-group ANOVA is *not* significant ($F = 0.07$, $p = 0.998$), while every modality-versus-baseline contrast is ($p < 0.001$); this confirms joint necessity but cannot rank importance. The coarse heads are more robust (token ~ 4 pp, type < 1 pp, parameter-type ~ 1.5 pp). Table III reports the per-modality numbers.

The frozen-encoder “joint necessity” is a gating artifact: the encoder routes command-discrimination signal through every modality at once, so zeroing any one degenerates the representation to one sufficient for the coarse heads but not command; a genuine ranking needs the retraining below.

1) *Per-Modality Encoder Retraining (Leave-One-Modality-Out)*: We retrain the encoder from scratch on the 98-feature base with each modality removed, train a fresh decoder on the



(a) Per-axis recoverability (5-fold mean): has-axis F1, motion-sign accuracy, and value MAE (inches). The Y-vs-Z sign asymmetry reflects the near-deterministic Y prior versus Z’s balanced sign distribution [15].



(b) The numeric-recovery “bathtub” (per-digit accuracy, 5-fold mean $\pm$ std, teacher-forced, patched retrain): slot 0 at 1.00 descends to the precision floor at slot 4 (0.42, thousandths, the finest tokenizer-retained digit); slot 5 returns to 0.99 as a structural zero of the V8 tokenizer’s 0.001 grid (discarding the ~ 3.31 -bit source digit), not information-bearing recovery [15]. Source: audit/patched_numeric_analysis.json.

Fig. 2: The numeric tier: per-axis summary (left) and per-digit decomposition (right).

TABLE III: Sensor-modality ablation (mean $\pm$ std; $n = 5$ folds per modality, except audio-RMS at $n = 4$ — one fold’s ablation run was unrecoverable). Each row zeros the named sensor channels at the encoder input and re-evaluates the decoder; the encoder is *not* retrained, so this is a lower bound on modality value. A one-way ANOVA across the seven zeroed-modality groups on command accuracy is not significant ($F = 0.07$, $p = 0.998$): the seven modalities are statistically indistinguishable from one another. Every modality versus the full-modality baseline *is* significant (Welch t , $p < 0.001$ in all seven cases), with a uniform ~ 40 – 44 pp command-accuracy collapse. Token, type, and parameter-type heads stay within ~ 4 pp of baseline for every removal.

Modality removed	Token	Type	Command	Param-type	Numeric
<i>baseline (5-fold)</i>	0.781 ± 0.022	0.984 ± 0.009	0.888 ± 0.056	0.993 ± 0.004	0.585 ± 0.033
zero accelerometer	0.743 ± 0.010	0.979 ± 0.010	0.484 ± 0.096	0.978 ± 0.002	0.471 ± 0.008
zero color	0.741 ± 0.010	0.977 ± 0.008	0.457 ± 0.108	0.977 ± 0.004	0.471 ± 0.009
zero electrical	0.741 ± 0.009	0.976 ± 0.009	0.449 ± 0.086	0.978 ± 0.002	0.465 ± 0.009
zero environmental	0.743 ± 0.010	0.977 ± 0.008	0.474 ± 0.077	0.978 ± 0.002	0.470 ± 0.011
zero gyroscope	0.741 ± 0.009	0.979 ± 0.005	0.462 ± 0.049	0.978 ± 0.003	0.469 ± 0.006
zero magnetometer	0.735 ± 0.018	0.975 ± 0.010	0.465 ± 0.084	0.978 ± 0.002	0.466 ± 0.017
zero rms	0.739 ± 0.010	0.978 ± 0.007	0.460 ± 0.088	0.978 ± 0.002	0.465 ± 0.010

The uniform command collapse with a non-significant between-modality ANOVA ($p = 0.998$) is the key result: the inference-time zeroing test confirms all modalities are jointly necessary but cannot rank them. The genuine modality-importance ranking is provided by the leave-one-modality-out encoder retraining (Section VI-G1).

resulting frozen encoder, and score all seven cells against a retrained full-stack baseline on a byte-identical held-out split. The retrained baseline reaches command accuracy 0.904 ± 0.039 , and every removal lands within ± 0.013 ; the fold-blocked ANOVA is non-significant ($F = 0.91$, $p = 0.499$) and the Holm-corrected paired test rejects none (0/7). Token and numeric heads behave the same. Figure 3 charts the paired deltas against the frozen collapse.

The -42 pp frozen-encoder collapse reduces to ≤ 1.3 pp under retraining (none significant), confirming the gating artifact: retraining reallocates capacity onto the remaining channels.

The deployment statement upgrades from “the full multi-modal stack is required” to “no single modality is individually load-bearing.” Two orthogonal probes corroborate this: per-board leave-one-sensor-out (LOSO) removals cost ≤ 2.6 pp (fold-blocked ANOVA $p = 0.415$; 0/6 Holm-significant) and the nested 6×6 (board $\times$ modality) interior is flat within ± 0.022 ($p = 0.896$); the redundancy is dense, not a marginal-average effect.

H. Cross-Class Generalization (Leave-One-Class-Out)

For each of the nine operation classes we train the decoder without that class and evaluate on its held-out samples, testing

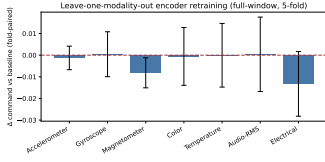


Fig. 3: Leave-one-modality-out encoder retraining: per-modality fold-paired Δ command vs. the retrained baseline; every modality within ± 0.013 , none Holm-significant (fold-blocked ANOVA $F = 0.91$, $p = 0.499$). Contrast the -42 pp uniform collapse under *frozen*-encoder inference-time zeroing (Table III): the gap is the encoder’s re-allocation capacity, not modality information content.

cross-operation generalization versus per-class memorization. Each holdout re-stratifies the remaining 8-class fold-1 partition at the headline 80/20 split (fold 1 is also the covariate-shift outlier of Section VIII, so the LOCO numbers are conditional on this partition choice); the coverage-repair step is deliberately *not* re-run, leaving the open-vocabulary residual exposed. The nine classes are the corpus population, so we report the finite-population descriptive distribution and make no wider inferential claim; each holdout is single-fold.

Across the nine holdouts, command accuracy collapses to mean 0.213 (min 0.000, median 0.21, max 0.678; $\sigma = 0.191$), token to 0.501, and numeric to 0.152; type and parameter-type survive (0.883 and 0.945). Relative to the in-distribution 5-fold baseline (0.888 command), this is a ~ 67 pp descriptive gap, not a two-sample test: file-stratified CV and class enumeration are different designs, and the per-cell command ranges do not overlap. Eight of nine holdouts cluster in $[0.000, 0.273]$; damagepocket reaches command 0.678 because it shares its predominantly-G1 motion vocabulary with the retained pocket variants, while damageadaptive (0.000) and damageface (0.094) do not. Cross-class generalization therefore succeeds only when the held-out class shares motion vocabulary with retained classes. The deployment-realistic claim is bounded by these LOCO numbers, not the 5-fold headline, and is scoped to closed-world audit over the operation classes seen at training time; open-world deployment is out of scope (Section VIII). Figure 4 charts the per-holdout, per-head breakdown.

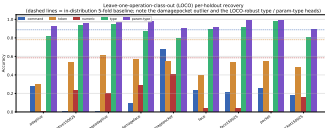


Fig. 4: Leave-one-class-out recovery per held-out class: command/token/numeric/type/parameter-type accuracy; dashed lines are the in-distribution 5-fold baselines. Eight of nine holdouts collapse on command (≤ 0.28); damagepocket (0.68) shares motion vocabulary with retained pocket variants. Type and parameter-type stay near baseline throughout.

I. From-Scratch Seq2Seq Baseline (Extended Version)

A from-scratch Transformer seq2seq baseline matches the headline teacher-forced command accuracy (0.877 ± 0.031 vs. 0.888 ± 0.056) and exceeds its teacher-forced numeric accuracy (0.754 vs. 0.585, an advantage that does not survive autoregression), but collapses to the most-frequent-class prior under AR; the extended version shows the apparent AR categorical gap reflects the conditioning-regime difference (reference-conditioned 0.888 vs. measured free-running 0.483, still above the baseline’s AR-derived 0.122), with the categorical heads carrying the audit-mode signal when a reference stream conditions the decoder.

VII. DISCUSSION

A. What is recoverable, and the class-prior boundary

Recoverability factorizes into three structural tiers (Section VI-B, Table II). Categorical structure (command and axis identity) is strongest: the sensor pathway fixes the motion class and the axes involved independently of file position, and positional metadata adds nothing (Section VI-F). Motion sign follows, with a prior-driven axis asymmetry (Y near-perfect, X and Z lower). Numeric precision is the limiting tier: a 64-second window observes relative motion, not absolute machine coordinates, so loss concentrates at the fine-value digits (Section VI-D).

a) *Is categorical recovery sensor-driven or operation-class structure?*: The frozen encoder was trained for nine-class operation classification on the *same* fold partitions used here, so the headline command accuracy could be operation-class structure re-expressed at the command level. A class-prior control (audit/confound_class_prior.json) tests this at every command-token position: the decoder reaches 0.795 ± 0.121 (5-fold) against an operation-class modal null of 0.572, the strictest legitimate history-free sensor-independent baseline since each test file is unseen. The decoder’s figure is measured under teacher forcing with ground-truth within-window history; a sensor-free per-class Markov-1 null granted the same history reaches 0.662 ± 0.004 (audit/markov_command_null.json), so the history-matched increment is $\approx +13$ pp. The 0.795 is measured only at command-token positions of the full-window predictions, a different position subset and aggregation than the 0.8875 head accuracy of Table I; the two are consistent measurements of the same predictions, not a discrepancy. The $+22$ pp increment is a 4-of-5-fold effect, not a uniform one: fold 1 (the covariate-shift outlier, Section VIII) is at or below its class prior, while folds 2–5 sit $+0.27$ to $+0.30$ above (paired-by-fold mean $+0.222$; bootstrap 95% CI $\approx [+0.10, +0.29]$). At $n = 5$ we report it as an effect size, not a p -value. Re-deriving the class-modal null on the test split instead of the training split does not shrink it (0.576 vs. 0.572), so roughly 0.57 of the 0.79 is prior structure and the decoder adds a within-class increment on top. That increment is an *upper* bound (closed-vocabulary regime, frozen-encoder representation reuse) but unchanged under the test-derived class-modal null.

B. Sensor-stack redundancy and deployment recommendations

a) *Sensor-modality importance.*: Under the frozen encoder, inference-time modality removal collapses command accuracy uniformly (between-modality $p = 0.998$), but encoder-retraining LOMO/LOSO ablations (Section VI-G1) show no single modality or sensor board is individually load-bearing: expected cost is ≤ 1.3 pp per modality and ≤ 2.6 pp per board. A cost-reduced post-hoc audit installation is therefore viable on this platform.

C. Security Implications and the Companion Threat-Model Study

The per-field recoverability characterized here bounds what a sensor-side integrity monitor can and cannot see: categorical substitutions (command, axis, motion-sign) leave a recoverable trace, while numeric-coordinate and feed-rate edits do not (Section VI-C). A companion security study (in preparation) develops the corresponding tamper-detection operating points, alert-budget analysis, and adaptive-adversary bounds under the same closed-world, single-platform scope; those results are reported separately so that this paper’s claims rest on recoverability alone.

VIII. LIMITATIONS AND FUTURE WORK

Every recoverability claim is conditional on one desktop CNC platform and one workpiece material; cross-platform replication on a production-class VMC is the prerequisite for any deployment claim and the principal open direction. Feed rate, spindle speed, and arc parameters have negligible corpus support, so their Table II accuracies are artifacts of near-zero baseline, not sensor-driven recovery. The 4 Hz stack aliases 77.7% of rows into a single sample period, capping numeric precision independently of the sequence head; a higher-rate stack with row-level encoder supervision is the main future direction. The file-level 5-fold splits are closed-vocabulary (every test line was seen in its fold’s training), and the open-vocabulary evidence is the leave-one-class-out protocol, under which command accuracy degrades from 0.888 to 0.213. A per-field covariate-shift audit flags fold 1 as the outlier fold (modest, < 6 pp shifts on low-support fields; `audit/extended_rigor_audits.json`), the basis for the fold-1 conditionality noted in Sections VI-B and VI-H.

IX. CONCLUSION

We characterize which components of an executing G-code block are recoverable from multi-modal CNC sensor data, using G-Code Decoder, a six-head Transformer decoder on a frozen multi-modal encoder with grammar-constrained inference under five-fold file-level cross-validation. Three findings follow. (1) **Per-block categorical recovery** (command, axis, motion direction) is sensor-driven under reference-conditioned evaluation (free-running it collapses to 0.483); command accuracy is 0.888, and the class-prior-controlled increment is unchanged under a test-derived class-modal null and $\approx +13$ pp against a history-matched Markov null (Section VII-A0a); on

an *unseen* operation class it collapses to the open-vocabulary floor of 0.213 (Section VI-H). (2) **End-to-end token and numeric reproduction is at floor**: token accuracy falls from 0.7811 (TF) to 0.2148 (AR), numeric from 0.5850 (TF) toward floor, and whole-sequence exact-match is 0.0263, bounded by exposure-bias mode-collapse and 4-Hz aliasing atop an information-theoretic limit the companion limit paper [15] characterizes. (3) **Sensor-stack redundancy** is dense ($\Delta \leq 2.6$ pp per board), enabling a cost-reduced post-hoc audit installation. The forensic-support post-hoc audit posture is the boundary within which these results hold, not real-time SOC alerting.

APPENDIX A SPECIFICATION ADDENDA

The coverage-repair procedure swaps files until every test line also appears in its fold’s training partition, converging in 0–3 swaps per fold (assignment released as `audit/file_fold_assignment.csv`). The grammar tokenizer quantizes each axis to a fixed per-axis precision (X/Y/Z at 10^{-3}) with sign carried by a separate head; empirical bucket counts sum to $K = 2,418$, and any out-of-range test index falls back to UNK (closed-vocabulary regime). Representation reuse is bounded two ways: class-label leakage caps the leakage component of command accuracy at 0.572, and a leave-decoder-test-files-out LOMO-baseline check bounds the empirical reuse effect at $|\Delta| \leq 0.016$ (0.904 ± 0.039 vs. headline 0.888).

APPENDIX B RELATIONSHIP TO THE COMPANION CONTRIBUTIONS AND DATA AVAILABILITY

This study is one of four related contributions: an encoder paper [10] establishing the multi-modal denoising sensor encoder and its 96.3% nine-class operation accuracy, a grammar paper [11] defining the hybrid G-code tokenization, the present per-row recoverability characterization, and a companion limit paper [15] deriving the decoder-agnostic information-theoretic and 4 Hz sub-Nyquist floor that bounds it. Beyond these four, a security companion (in preparation) develops the deferred tamper-detection operating points, and an extended journal version reports the full ablation matrix. The 120-file corpus, per-fold arrays, prediction JSONs, and audit artifacts are released (Data Availability Statement); nothing required to reproduce these results depends on the companion contributions.

APPENDIX C REPRODUCIBILITY

Training uses AdamW ($\eta = 5 \times 10^{-5}$, weight decay 0.05, batch 64, up to 300 epochs, patience 75) at seed 42; the headline command accuracy is seed-robust (four-seed 5-fold means within a 0.0085 span, far inside the ± 0.0558 cross-fold spread; `audit/seed_robustness_full_window.json`). The full hyperparameter grid, per-modality retraining protocol, and statistical exhibits are in the released artifacts (Data Availability).

DATA AVAILABILITY

Sensor datasets, analysis code, per-fold splits, checkpoints, and every cited audit/*.json artifact are publicly available at https://github.com/stephenseacuello/gcode_fingerprinting (release v1.0-submission; Zenodo DOI in the repository README; code MIT, corpus CC BY 4.0).

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